

ÉTIENNE MITCHELL-BOUCHARD

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EDUCATION

Professional Master's in Machine Learning, Computer Science

MILA & Université de Montréal

September 2025 - April 2027

Montréal, QC

Bachelor's in Computer Science

Université de Montréal

September 2022 - April 2025

Montréal, QC

- Recipient of the Computer Science Excellence Scholarship from the Ministry of Higher Education

DEC in Science, Computer Science and Mathematics

Collège Lionel-Groulx

August 2020 - May 2022

St-Thérèse, QC

EXPERIENCE

Teaching Assistant

Université de Montréal

September 2024 - April 2025

Montréal, QC

• Courses :

- [IFT1119](#) – Intro to Video Game Programming – Fall 2024, Fall 2025
- [IFT2200](#) – Video Game Programming – Winter 2025
- [IFT1810](#) – Introduction to Programming – Winter 2025
- [IFT2720](#) – Introduction to Multimedia – Fall 2024, Winter 2025, Fall 2025
- [IFT1986](#) – Text Processing – Winter 2025, Fall 2025
- [IFT1901](#) – Introduction to computing tools – Fall 2025
- Graded over 500 student assignments
- Helped 250+ students understand course content
- Supervised midterms and final exams

Grocery Clerk

Maxi Fabreville

July 2020 - January 2025

Laval, QC

- Operated a forklift
- Restocked shelves
- Received deliveries
- Closed and locked up the store
- Managed the store during closing hours
- Managed staff and customers

TECHNICAL SKILLS

Languages: Python, C#, Java, HTML/CSS, JavaScript, C++, C, Bash, Haskell

Development Tools: Git, Unity, VS Code, VS, PyCharm, Jupyter, Google Collab, IntelliJ, CLion, WebStorm

Libraries/Frameworks: Tensorflow, Pytorch, Scikit-Learn, Numpy, Pandas, JavaFX, three.js

Other: UML, SQL, Google Cloud, Data Structures, Algorithms

PROJECTS

[tp-map](#) | *HTML, CSS, JavaScript*

[Github Repo](#)

- Interactive map that shows players the location of all items in the game
- No interactive map existed for Zelda: Twilight Princess despite high demand following the release of the Randomizer
- Used by 30+ users daily
- Management of a Discord server where users provide bug reports and suggestions

[matrix-robot](#) | *HTML, CSS, JavaScript*

[Github Repo](#)

- Animated robot using only matrix operations
- Utilized scaling, rotation, and translation matrices
- Robot has walking animation and can look around the scene
- Limbs move independently within realistic joint constraints

[tree-render](#) | *HTML, CSS, JavaScript*

[Github Repo](#)

- Scene rendering with a tree, leaves, apples, and shadows
- Multiple scenes with tree skeletons to show structure
- Used splines and meshes to create realistic branches

[OnlineShop](#) | *HTML, CSS, JavaScript, jQuery, AJAX*

[Github Repo](#)

- Interactive e-commerce front-end
- Real-time updates without page reloads
- Full order system for user interactions

[Space Race Unlimited](#) | *Unity, C#*

[Github Repo](#)

- Complete 3D spaceship racing game
- LAN multiplayer supported
- High score tracking

[Zelda II Clone](#) | *Unity, C#*

[Github Repo](#)

- 2D action platformer prototype
- Character animations
- Basic AI-controlled monsters
- Combat system

[AI Price Predictor](#) | *Python, Jupyter, Scikit-Learn*

- Used various AI models to predict future prices of stocks and cryptocurrencies
- Models analyze values derived from past price data
- Final model (MLPClassifier) predicts price direction with 50–60% accuracy

[Trading Algo](#) | *Python, Binance's API*

- Autonomous cryptocurrency trading bot operating 24/7
- Follows a strategic algorithm for long-term gains
- Logs key market and trade data

[UniShop](#) | *Java, Javadoc, UML*

- Command-line based online store simulator
- Fully developed using Agile methodology
- UCSV-based database for users and product listings
- Includes various user interactions

[SQL-JavaFX](#) | *Java, JavaFX, PostgreSQL*

[Github Repo](#)

- Application for querying SQL databases
- Used JPA to map SQL tables to Java classes
- Supports custom SQL queries
- Includes basic query templates

[UdeM Registration System](#) | *Java, JavaFX*

[Github Repo](#)

- Course registration software for Université de Montréal
- Server handles requests
- Client-side graphical interface
- CSV database for students and course enrollment

[Natural Language Processor](#) | *Java*

- Custom optimized HashMaps for natural language analysis
- Features word search in files, autocomplete predictor, and spell checker

[Slip Programming Language](#) | *Haskell*

- Interpreter for a fully functional programming language similar to Lisp
- Includes lexical and syntax analysis, compilation, expression evaluation, type checking, and syntactic sugar